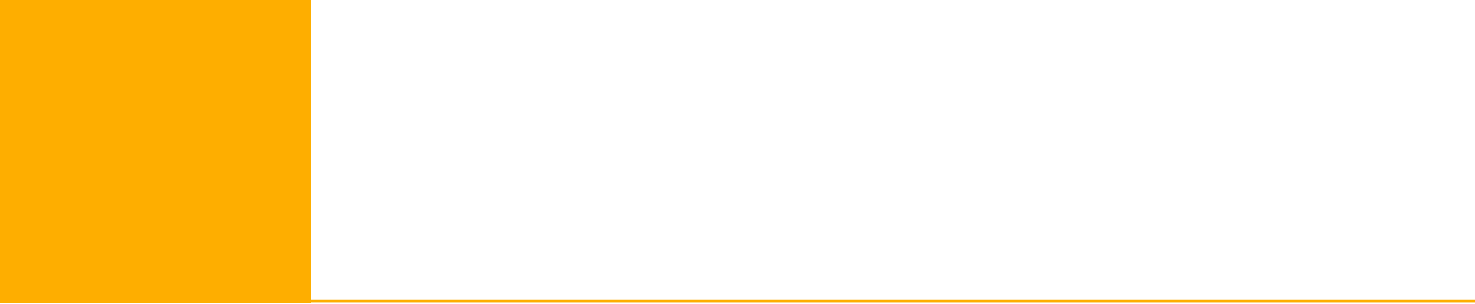


A solid orange horizontal bar at the top of the page, with a vertical orange line extending from the top left corner down to the bar.

FIT2179 Data Visualisation
Course Notes

TYPOGRAPHY

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TYPOGRAPHY

Typography is the technique and art of arranging type to display text and other visual elements to convey meaning in a legible and pleasing way.

A typeface influences how easy information is to read and how it is perceived, for example, how its trustworthiness is judged. Selecting a typeface is only one aspect that influences readability and perception of a visualisation or text. We also need to consider the spacing of lines (leading), the visual contrast with the background, alignment of text, and many other factors.

For data visualisation, we need to consider how we encode data attributes with type-related visual channels, such as font size and font weight, how we create a visual hierarchy to guide the user through a visualisation, and how we place labels to unambiguously annotate features on maps and diagrams.

There is something to be said for aesthetics: Aesthetic pleasure is an important aspect of any visualization. Aesthetics affect the usability and effectiveness of a visualisation and have the potential to engage viewers. If type is ugly or difficult to read, then readers will likely not spend much time with a visualisation, or ignore it altogether.

The latin script (also called Roman script) is the writing script used for English and many other languages. A large number of other scripts, such as Arabic, Cyrillic, Chinese, or diverse Indic scripts exist. Not all concepts and principles discussed in this note transfer to these other scripts.

THE GEOMETRY OF TYPE

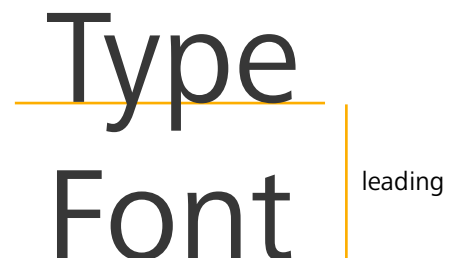
The **baseline** is the invisible line on which the glyphs rest. It is used to measure vertical distances.



The **x-height** is the distance between the baseline and the imaginary line at the top of the lowercase x. The x-height defines how high lowercase letters without ascenders (x, m, n, o, etc.) are compared to uppercase letters. Typefaces with tall x-heights tend to have better legibility at small font sizes.



Leading is the space between adjacent lines of type. It can be defined as the distance between two baselines.



Typography measures dimensions in **points**. One point is 0.353 mm ($1/72$ of an inch).

TYPOGRAPHIC TERMINOLOGY

TEXT

Text is what we read or write.

CHARACTER

A character is the smallest unit of text, representing a letter, number, punctuation mark or another typographic symbol.

GLYPH

A glyph is a visual symbol representing a character. Some glyphs represent multiple characters.

CHARACTER ENCODING

A character encoding matches a character to a glyph.



STYLE

A style consists of a set of glyphs with a consistent visual appearance. Common styles are thin, regular, bold, italic or condensed.

FONT

A font is a set of letters (or glyphs) in a particular style usable for typesetting. In digital typography, a font is stored in a font file.

TYPEFACE

A typeface combines various styles. A typeface is a set of fonts with a visually coherent design.

TYPOGRAPHY

Typography is the technique and art of arranging type to display text and other visual elements to convey meaning in a legible and pleasing way.



TERMINOLOGY

The terms **font** and **typeface** are closely related and sometimes used interchangeably, for example, Frutiger bold italic is both a typeface and a font. But font and typeface are not synonyms: *A typeface is what you see, and a font is what you use.* For example, when discussing the form of letters, we refer to a *typeface*, but a web page loads and uses a *font* file to display text.

TYPES OF TYPEFACES

The thousands of typefaces with diverse design traits are difficult to classify. Here are four basic classes: serif, sans serif, script and display. Each class is illustrated with typical typefaces.

Serif typefaces can be traced back to first lead type made for printing in the fifteenth century. Sans serif typefaces are widely used since the 1950s.

SERIF FONTS

Serifs are small strokes and other shapes attached to letter forms. Serif fonts have a classical look.

Serif

Bodoni 72

Serif type is considered traditional and sometimes old-fashioned. Serif type is used for printed books, magazines and newspapers. An example is this **Times New Roman**.

Didot and the contemporary Bodoni are elegant typefaces with high contrast between horizontal and vertical lines. They are used by fashion magazines, and cosmetic and luxury brands.

Garamond is a serif typeface for books with good readability. It looks smart and classic.

SANS SERIF FONTS

"Sans" means "without" in French, so sans serif fonts have no serifs. The letterforms are simple, even, and without decoration.

Sans

Frutiger

Most data visualisations use sans serif fonts, because they are considered clean and modern. An example is the **Frutiger** typeface used here.

Helvetica is a widely used typeface that is minimal and neutral. It influenced many derived typefaces, but long texts are not easy to read.

Open Sans has open glyph shapes and a large x-height, which results in good readability on screen and at small sizes.

Script and display typefaces are designed for titles, headlines and short sentences that should stand out. They can create a specific aesthetic, but their appearance must match the overall topic.

SCRIPT FONTS

Script fonts imitate handwritten letterforms. The styles vary from casual handwriting to formal calligraphic lettering.

Script

Snell Roundhand

Script fonts can be formal.
They can be childish.
They can be personal.
Or they can invite to a party.

Blackletter, also known as Gothic script, is a group of script typefaces developed in Western Europe. Today, blackletter is only used for decorative typesetting. For example, a number of newspapers, such as the New York Times, print their name on the masthead with a blackletter typeface. And, heavy metal bands like blackletter.

DISPLAY FONTS

Display fonts have eye-catching letterforms and are designed for large headings. They have poor legibility for long texts.

Display

Impact

Display fonts exist in a wide variety of styles. For example, they can be **geometric**, **ABSTRACT**, or **playful**.

Display fonts can strongly influence the appearance of a visualisation and evoke different associations. For example, film posters use specific display fonts for **WESTERN**, **HORROR**, or **animated** films.

Display fonts often only have a single style and a single weight and a reduced set of glyphs.

WEIGHT, STYLES, UPPERCASE AND COLOUR

TYPE FAMILIES

A type family consists of multiple styles. A basic set includes *regular*, *italic*, **bold** and **bold italic** styles. Type families can contain additional weights, combined with condensed and extended styles in any combination.

This text uses a Frutiger type family with more than 20 different styles.

The **weight** describes how thin or thick the strokes of a typeface are. Professional typeface families include a set of graduated weights. Very thin and very thick forms are more difficult to read than regular or normal weights that are designed for best readability of long text.

Abc Abc Abc Abc Abc Abc Abc
Ultra Light Light Regular Medium Bold Heavy Black
Frutiger.

Italic styles slightly slant letter forms to emphasise important words, indicate quotations or words from other languages. The more common upright style is called *regular* or *roman*.

Abc Abc Abc Abc Abc Abc Abc
Italic Frutiger with the same weights as above.

Condensed styles take up less space horizontally. Condensed styles are useful for long headings, tables, or maps and other data visualisations where space is limited and a large number of annotations or labels are added. A condensed style has reduced legibility compared to a regular style. Extended styles take up more space without improving legibility.

Abc Abc Abc Abc Abc Abc Abc
Condensed Frutiger with same weights as above.

TERMINOLOGY

Terminology for typographic weight and form variations is not standardised.

Typical terms for **weight** are *light* or *thin* for thin strokes, *regular*, *roman* or *book* for weights designed for reading long texts, and *medium*, *bold*, *heavy*, and *black* for weights with increasingly wider strokes.

Italic is also called *oblique* or *slanted*.

Condensed is also called *narrow* or *compressed*.

Extended is also called *wide* or *expanded*.

Some type families prepend *extra* or *ultra* to indicate styles that are *ultralight*, *extrabold*, *extra condensed*, etc.

UPPERCASE can be useful to emphasise words, for example, in titles, or on maps to distinguish important features, such as large cities, from less important features, such as small towns.

ABC Abc
Uppercase and mixed case.

The **colour** of type can be varied to highlight important information. For data visualisation, coloured words can replace a legend, or visualise a categorical attribute. Colour must be saturated and have a strong contrast with the background for good readability.

Abc Abc Abc Abc Abc Abc Abc

READABILITY OF TEXT

READABILITY AND LEGIBILITY

A goal of typography is to make text and graphic information easy to read. But what makes text easy to read?

TERMINOLOGY

Readability is the ease with which text can be read and understood.

Legibility is the ease with which glyphs can be recognized.

Legibility is a precondition for readability. If the glyphs are not legible or decodable, the reader is unable to read and understand the information in the text. Letters and words have good legibility when decoding is effortless and fast. For letters and words to be legible, the reader must be able to easily perceive them.

WHAT MAKES A LEGIBLE TYPEFACE?

A legible typeface uses generic letter forms that are familiar to the reader. The glyphs of an ornamental or experimental typeface are likely difficult to recognise and decode, and therefore have poor legibility.

Glyphs also need to be easy to distinguish from each other. For example, the lower case *a* and *o*, or *c*, *e* and *o* letters are difficult to distinguish with some typefaces.

WHAT INFLUENCES READABILITY?

Readability is influenced by legibility, but also the complexity and style of the text, the reading skills and expertise of the reader, as well as the context of the reader. That is, the quality of the illumination, electronic display, viewing distance, and other factors that can support or distract from reading.

LEGIBLE TYPEFACES

Early serif typefaces, such as Garamond and Bembo, were designed for books. They tend to be less fatiguing for long texts than sans serif typefaces. Short texts, such as letters, explanations for visualisations, road signs, labels on maps, or legends and annotations on charts are commonly set with sans serif typefaces that have been optimised for fast reading of short text. A good example is the Frutiger type family used for this text.

ROW LENGTH

If a line of text is too long, the reader has difficulty finding the start of the next line. For good readability, a line should not contain more than approximately 60 characters.

FONT SIZE AND LEADING

For print, a font size of 9 or 10 points is recommendable. On digital displays and for typefaces with a small x-height, the font size should be 11 or 12 points for good legibility.

9 points

10 points

12 points

Leading is often 120% of the font size.

More leading can increase legibility with long rows and fonts with large x-heights.

CONTRAST

A strong contrast between text and background is required for good legibility. Contrast can be increased with heavier weights and contrasting colour.

Poor

Poor

Good

Good

Queen Hera
or
Queen Hera

ao

Futura

ceo

Helvetica

READABILITY AND LAYOUT

This column is too wide and leading (the space between adjacent lines) is much too small. This results in a block of text with extremely poor readability. The amount of required leading depends on the vertical extent of the glyphs of a typeface and the number of words on a line. With too little leading (as with this text), the lower end of descenders of characters like y, p, or g can intersect with the upper end of ascenders of characters like f, l, or k. For legible text, this must be avoided. Also, if a line is too long (as with this text), the reader's eye has to travel a long distance from the end of the current line to the beginning of the next line, and it is likely that the reader continues on the wrong line. Additional leading increases readability by reducing the chance of this happening. So how many words should there be on a line to make it easy for the reader's eye to jump from one line to the next line? Adjusting the width of a text block such that there are between seven and ten words on a line generally results in good readability. In general, a longer line with many words requires more leading than a short line with few words.

This text is set with the same 9-point typeface as the text above, but the column is much narrower. This increases readability, but leading has not been sufficiently increased. Overall, readability of this short text is still poor.

This text is set with the same 9-point typeface and the same column width as the text above, but the leading has been extended considerably. This results in a block of text that is easy to read. One line of text has approximately seven to ten words. As a rule of thumb, a row should have about 60 characters.

This text is set with the same 9-point font and leading as the text above, but it is grey instead of black. This reduces the contrast with the background and makes it quite difficult to read, especially on a background that reduces contrast.

Certain color combinations can be unpleasant to the eye and are difficult to read, for example this

Center-aligned text is difficult to read across multiple lines. Left-aligned text should be preferred for text that spans multiple lines, because left alignment simplifies navigating from line to line and often results in a more balanced design.

Justified text, such as this paragraph, adjusts word-spacing and optionally letter spacing so that words align with the left and right margins. Justified text can result in irregular white space between words, especially when a line contains only a few words. Justified text does not increase the readability of text, but can help create a balanced layout.

TYPOGRAPHY FOR IMPROVING READABILITY

Indicate ranges with an en-dash (for example, 23–456) or the word “to” (456 to 789) instead of a hyphen (23-456 😞).

Never underline text for highlighting information; underlines are a relict from typewriters and only useful to indicate web links.

A space separates numerals and units (for example, 12.3 cm, 456 kg). The percentage sign (15%) and the degree sign (360°, 37°C) are not separated by a space.

ENCODING DATA ATTRIBUTES WITH TYPE

The size and weight of text can be varied to visualise ordered data attributes. Italic, upper case and colour hue can be varied to show different categorical data attributes.

SIZE

Varying the size of words uses the size channel, so varying type size is appropriate for visualising ordered data attributes.

WEIGHT

Varying the weight of type can also visualise an ordered attribute. More ink indicates a greater attribute value. For example, a bold label on a map indicates a larger or more important town than a thin label.

ITALIC

The combination of Italic and regular type can indicate a categorical attribute with two possible values. For example, on maps, italic styles are often used to indicate mountain peaks or passes, and regular styles are used to indicate populated places.

UPPERCASE AND MIXED CASE

Uppercase can replace mixed case to indicate the variation of a categorical attribute with two possible values. For example, maps can label capitals entirely with uppercase letters and other locations with mixed case.

COLOUR HUE

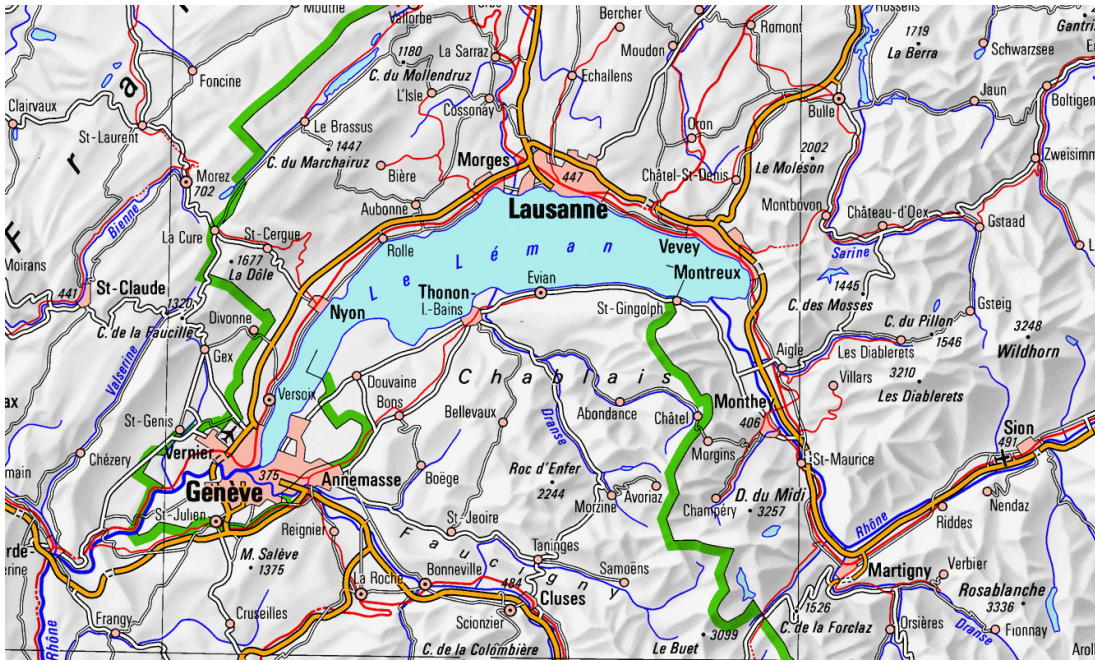
Varying colour hue can visualise a categorical data attribute. Varying brightness and saturation of text is not effective, because such variations would result in weak colour contrast with the background colour and reduce readability of the text.

COMBINATIONS

It is possible to combine variations of size, weight, italic style, upper case and hue for visualising data attributes. This is useful for visual emphasis. For example, the size and weight can both increase or decrease for an ordered attribute, or italic/regular and hue can both vary for categorical attributes.

Size	New York	Cairns	Geelong	Clayton	Ordered
Weight	Pacific Ocean	Tasman Sea	Bass Strait	Port Philip Bay	Ordered
Italic		<i>Mt Everest</i>	Kathmandu		Categorical
Case		CANBERRA	Sydney		Categorical
Colour hue	Lake Wellington	Victoria	Yarra Ranges National Park		Categorical

ENCODING DATA ATTRIBUTES: AN EXAMPLE



This map of the Lake Geneva area varies and combines typographic styles, weights, font sizes and colours to show ordered and categorical attributes. It uses a single type family, the sans serif Univers, mainly in a condensed style to save space.

Cossonay

Towns and villages vary point size and weight with population size; the smallest towns use a tiny condensed type, whereas Genève is set with a large and bold font.

Morges

Lausanne

Mountain peaks and mountain passes are set in condensed italic to distinguish them from towns, for example, *Les Diablerets*.

Lake Geneva (*Le Léman*) and rivers combine dark blue with an italic style.

Labels for regions are set in italic and they vary in size. For example, *France* in the top-left corner is much larger than *Chablais*, which is south of Lake Geneva. Labels for regions also add additional space between individual glyphs, such that the label covers the entire region on the map. For example, *Chablais* becomes *C h a b l a i s*.

This map does set any labels entirely in uppercase letters, which is common to indicate capitals on maps.

And this last

**You will read
this first**

And then you will read this
Then this

And this last

VISUAL HIERARCHY WITH TYPOGRAPHY

You will read this first

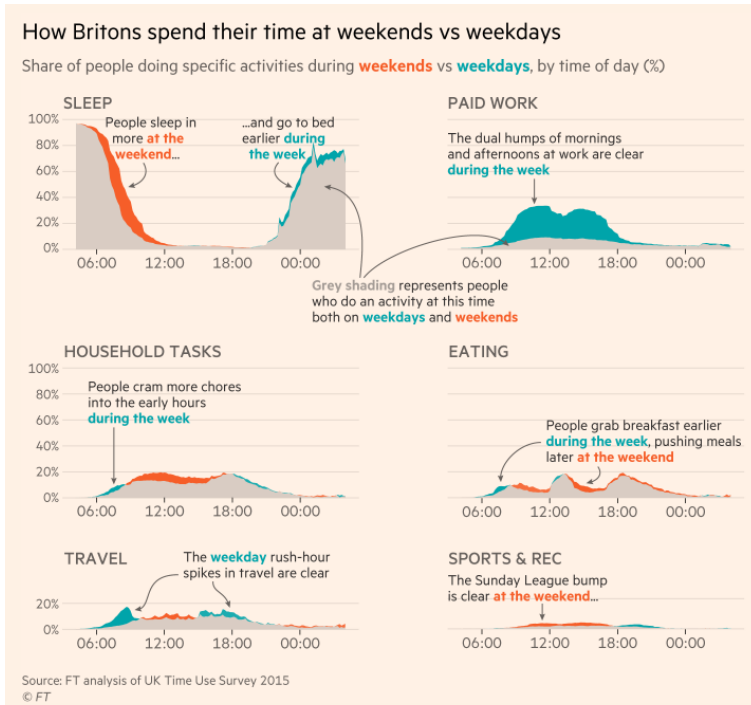
①

Weight, size and colour contrast are effective means to guide the eye of the viewer through a page. The human eye is attracted to high-contrast text ① – which is used for the large central label – and it then scans the page in downwards direction, from left to right, until it finds the second ② and third ③ labels. They are smaller and grey, and placed on a darker background, which reduces visual contrast. It is only when we have reached the bottom of the page, that our eye perhaps discovers the fourth ④ small label at the top-left.

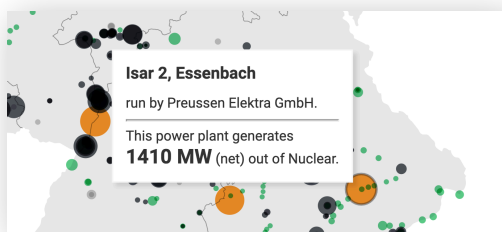
② And then you will read this

③ Then this one

VISUAL HIERARCHY WITH TYPOGRAPHY



This visualisation makes good use of typography to create a visual hierarchy that guides the viewer's eye. The title is set with a relatively large font size in black colour to create a strong visual contrast to attract the eye from the start. The subtitle is smaller and grey, but includes coloured words to explain the colour coding of two categorical attribute values. The colour coding makes a legend explaining the orange and teal colours unnecessary. Each of the six diagrams is labeled with an all-uppercase word that is bold and enlarged. The axis labels are grey and relatively small, because they convey secondary information and are repeated across the six diagrams. In contrast, the annotations are black to stand out, and again use colour-coded words to explain the meaning of the hue channel.



Typography for this tooltip of an interactive map also creates a clear visual hierarchy of the displayed information. The most important information – the name of the hovered power plant and its net production – are highlighted with bold, high-contrast and enlarged text.

LICENSING FONTS

Fonts, like other digital goods, can be duplicated and shared. Fonts are licensed similar to computer software, so being aware of the terms and conditions is important. The end-user license agreement defines how a font can be used, for example, for print, web, apps, or e-books and the maximum number of users.

Google Fonts provides an easy-to-use collection of many high-quality free fonts. When used on a webpage it comes at the price of sharing the client's IP address with Google.